Name: ……**Sign** ……..**Comb**……..

P530/1

BIOLOGY

THEORY

UGANDA ADVANCED CERTIFICATE OF EDUCATION

KIGEZI HIGH SCHOOL

S.6 BIOLOGY MID TERM ONE 2026

Paper one

- ***Answer all the items in this paper.***
- ***Organise your answers, then present them logically and precisely.***

SECTION A

Answer all the items in this section.

Item 1

Victoria Nile is a source of food and income for the communities living along it. However, the community near a textile factory have reported to the district authorities that the fish catches have significantly reduced over the past few months. The authorities requested Dr. Achen, a fisheries biologist from National Environmental Management Authority (NEMA) to carry out an investigation. She suspected that the cause was pollution from the factory that releases wastes containing heavy metals and cyanide into the lake.

She caged fish at two different sites; a clean control site (no pollution) and a site near the factory outflow. She investigated the effects of the heavy metals on the gill and mitochondrial structures and ATP synthesis. The data she collected is provided in table 1.

Table 1

Fish group	ATP Concentration (µmol/g cell)	Thickness of gill squamous cells	Number of ribosomes per mitochondrion ($\times 10^6$)	Concentration of oxidised NAD(µmol/g tissue)	Space between cells gill epithelium	Amount of cyanide in river water (mg/L)	Amount of lead (Pb) in river water (mg/L)
Control (no pollution)	9.5	Thin	9.8	7.2	0.5µm	0.01	0.01
Polluted site – factory outflow	2.2	Thick	3.5	1.3	6.0µm	1.00	1.00

You are tasked with analysing her findings and proposing a course of action.

Task

Analyse how cellular and tissue changes in the fish from the polluted site affect respiration and result in reduced fish catches to propose and justify a sustainable management plan to ensure health of organisms in the river.

Item 2

Napak district is one of the agricultural areas in Uganda. It is a lowland area with high temperature, shallow soils that have limited organic matter and very low moisture. Farmers grow both native and non-native crops, however, they often get low yields from the non-native crops unlike from the native crops.

The farmers invited agricultural extension workers to study the situation and propose strategies to improve the yields. They conducted a study in the district and the data collected on native and non-native crops grown is shown in table 2.

Table 2

Condition	Native plants	Non-native crops grown
The type of leaf anatomy	Kranz anatomy present	Kranz anatomy absent
Distribution of	Many stomata on lower	Many stomata on upper

stomata	leaf epidermis	leaf epidermis
Size of leaf cuticle	Thick	Thin
Number of hairs on the leaf	Numerous	Very few

You are part of the team invited by the agricultural extension officer to analyse the data and propose strategies to the farmers on how to improve their crop yields.

Task:

(a) Account for the difference in the yields between native and non-native crops grown in the district, and propose strategies to enhance productivity and resilience.

SECTION B

Item 3

At a research laboratory of the National Agricultural Research Organisation in Kampala, scientists are developing maize varieties resistant to stem borers using gene technology techniques. The resistant gene is to be extracted from a wild plant variety and then inserted into the maize plants. When genetically modified maize plants were crossed with normal ones, all maize plants produced in first generation were resistant. In the second generation after interbreeding the produced maize plants, some maize plants were not resistant to the stem borers. The garden had a total number of 600 maize plants in the second generation

Task

Analyse the above information and explain how the genetically modified maize was formed, ethical, social, and environmental implications of such activities, also predict the number of resistant plants in the second generation, and evaluate a possible method to tell the genetic composition of the maize plant showing dominant characteristics.

Item 4

Communities living around Echuya Forest Reserve in south-western Uganda have reported increasing invasions of native rodent species into their crop gardens.

The National Forestry Authority conducted an investigation to determine the cause of the problem. The investigation found that the population of native rodents within the forest had declined, while that of a recently introduced non-native rodent species had increased rapidly. Field observations showed that the non-native rodents possess stronger limbs, larger incisors, reproduce more frequently, and grow faster after feeding compared to the native rodents. Researchers also noted that farm chemicals from nearby agricultural areas enter parts of the forest through runoff during heavy rains.

Samples collected from different components of the forest ecosystem revealed the following results:

Component of ecosystem	**Average concentration of farm chemicals (ppm)**
Forest plants	0.5
Insects feeding on plants	1.2
Rodents feeding on insects and plants	3.8
Birds of prey feeding on rodents	7.6

In addition, feeding experiments conducted by researchers showed the following results:

Rodent species	**Average food consumed per week (g)**	**Average number of rodents that are affected by the pesticide**
Native rodent species	50	45
Non-native rodent species	200	5

Crop damage reported by farmers around the forest increased from **12% in 2019 to 38% in 2024**.

Task

Using the information provided, analyse the data to explain the possible ecological reasons for the displacement of native rodents into crop gardens and

the increasing dominance of the non-native species, and propose a **practical conservation and management plan** that could help restore balance in the ecosystem while protecting community crops.

END;

THE DIFFERENCE LIES IN OUR MOTIVATION AND PRIORITY